

F26 AerE 3310: Flight Control Systems I

Practice Problem Set 3: IVT, FVT and Transfer Functions

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Problem 1

For each of the following, the Laplace transform $F(s) = \mathcal{L}[f(t)]$. Determine the correct value of the limit and explain if we can apply initial or final value theorem (IVT or FVT) in each case.

(a) $\lim_{t \downarrow 0} f(t)$ for $F(s) = \frac{s^3 + 2s^2 + 4s + 7}{s^3 + 2s}$

(b) $\lim_{t \uparrow \infty} f(t)$ for $F(s) = \frac{3}{s(s-2)}$

(c) $\lim_{t \uparrow \infty} f(t)$ for $F(s) = \frac{1}{(s+5)(s^2+s+1)}$

(d) $\lim_{t \uparrow \infty} f(t)$ for $F(s) = \frac{1}{s^2 + \omega^2}$ where $\omega \in \mathbb{R}$

(e) $\lim_{t \uparrow \infty} f(t)$ for $F(s) = \frac{k}{s}$ where $k \in \mathbb{R}$

Solution.

(a) Cannot apply IVT because $F(s)$ is not strictly proper. To determine the correct value of the limit, we perform partial fraction expansion:

$$\begin{aligned} F(s) &= \frac{s^3 + 2s^2 + 4s + 7}{s^3 + 2s} \\ &= \frac{s^3 + 2s}{s^3 + 2s} + \frac{2s^2 + 2s + 7}{s^3 + 2s} \\ &= 1 + \frac{A}{s} + \frac{Bs + C}{s^2 + 2} \end{aligned}$$

which gives

$$2s^2 + 2s + 7 = A(s^2 + 2) + Bs^2 + Cs.$$

Evaluating the above at $s = 0$ gives $A = 7/2$, at $s = 1$ gives $B + C = 1/2$, at $s = -1$ gives

$B - C = -7/2$, i.e., $B = -3/2$, $C = 2$. Hence,

$$\begin{aligned} f(t) &= \mathcal{L}^{-1}[1] + \frac{7}{2}\mathcal{L}^{-1}\left[\frac{1}{s}\right] - \frac{3}{2}\mathcal{L}^{-1}\left[\frac{s}{s^2+2}\right] + 2\mathcal{L}^{-1}\left[\frac{1}{s^2+2}\right] \\ &= \delta(t) + \frac{7}{2} \cdot 1 - \frac{3}{2} \cos(\sqrt{2}t) + \frac{2}{\sqrt{2}} \sin(\sqrt{2}t). \end{aligned}$$

Therefore, $\lim_{t \downarrow 0} f(t) = \infty + \frac{7}{2} - \frac{3}{2} + 0 = \infty$.

(b) Cannot apply FVT because of the right-half-plane (RHP) pole $s = 2$. To determine the correct value of the limit, we perform partial fraction expansion:

$$\frac{3}{s(s-2)} = \frac{A}{s} + \frac{B}{s-2} \quad \Leftrightarrow \quad 3 = A(s-2) + Bs,$$

which upon evaluating at $s = 0$ gives $A = -3/2$, and at $s = 2$ gives $B = 3/2$. Hence,

$$f(t) = -\frac{3}{2}\mathcal{L}^{-1}\left[\frac{1}{s}\right] + \frac{3}{2}\mathcal{L}^{-1}\left[\frac{1}{s-2}\right] = -\frac{3}{2} + \frac{3}{2}e^{2t}.$$

Therefore, $\lim_{t \uparrow \infty} f(t) = -\frac{3}{2} + \infty = \infty$.

(c) Can apply FVT because all three poles $s = -5, -\frac{1}{2} \pm j\frac{\sqrt{3}}{2}$, are in the open left-half-plane (LHP). Therefore,

$$\lim_{t \uparrow \infty} f(t) = \lim_{s \rightarrow 0} sF(s) = \frac{s}{(s+5)(s^2+s+1)} = \frac{0}{5 \cdot 1} = 0.$$

(d) Cannot apply FVT because both poles $s = \pm j\omega$ are in the closed RHP. Since

$$f(t) = \mathcal{L}^{-1}\left[\frac{1}{s^2 + \omega^2}\right] = \frac{1}{\omega} \sin(\omega t),$$

therefore $\lim_{t \uparrow \infty} f(t)$ does not exist.

(e) Can apply FVT because the pole at the origin is not repeated, and is the only pole. Therefore,

$$\lim_{t \uparrow \infty} f(t) = \lim_{s \rightarrow 0} sF(s) = \lim_{s \rightarrow 0} s \cdot \frac{k}{s} = \lim_{s \rightarrow 0} k = k.$$

Problem 2

Consider the equation of motion for the simple pendulum linearized about $(x_1, x_2) := (\theta, \dot{\theta}) = (0, 0)$ that we derived in Lec. 5 p. 1-3 in LTI state space form:

$$\begin{pmatrix} \dot{x}_1 \\ \dot{x}_2 \end{pmatrix} = \begin{bmatrix} 0 & 1 \\ -\frac{g}{\ell} & 0 \end{bmatrix} \begin{pmatrix} x_1 \\ x_2 \end{pmatrix} + \begin{pmatrix} 0 \\ \frac{1}{m\ell^2} \end{pmatrix} u. \quad (1)$$

Consider initial conditions: $x_1(t=0) = 0$, $x_2(t=0) = 0$. Here, u is the input torque provided by a motor. Suppose that we are interested in the angle as the only output, that is, $y = x_1$.

(a) By taking component-by-component Laplace transform of (1) and eliminating appropriate variable, determine the transfer function $\frac{Y(s)}{U(s)}$. Show all your calculations.

Solution for Problem 1(a)

Taking component-by-component Laplace transform of the state equation (1), and using the given initial conditions, we obtain

$$sX_1(s) = X_2(s), \quad sX_2(s) = -\frac{g}{\ell}X_1(s) + \frac{1}{m\ell^2}U(s). \quad (2)$$

Taking the Laplace transform to both sides of the given output equation $y = x_1$, we get $Y(s) = X_1(s)$. Therefore, the transfer function

$$\frac{Y(s)}{U(s)} = \frac{X_1(s)}{U(s)}. \quad (3)$$

To eliminate $X_2(s)$ in (2), we compute

$$s^2X_1(s) = sX_2(s) = -\frac{g}{\ell}X_1(s) + \frac{1}{m\ell^2}U(s) \quad \Rightarrow \quad \left(s^2 + \frac{g}{\ell}\right)X_1(s) = \frac{1}{m\ell^2}U(s),$$

which combined with (3) yields the desired transfer function

$$\frac{Y(s)}{U(s)} = \frac{1}{m\ell^2} \frac{1}{s^2 + \frac{g}{\ell}}. \quad (4)$$

(b) Argue in 1 sentence whether the transfer function derived in part (a) is proper or strictly proper. Compute all poles of the transfer function you derived in part (a).

Solution for Problem 1(b)

Strictly proper because for the transfer function $\frac{Y(s)}{U(s)}$ has degree of numerator polynomial (0) < degree of denominator polynomial (2).

The poles are the roots of the denominator polynomial of the transfer function, which in this case are pure imaginary: $s = \pm j\sqrt{g/\ell}$ where $j := \sqrt{-1}$.

(c) Use your transfer function in part (a) to derive $y_u(t)$ which is the output for the unit step input

$$u(t) = \begin{cases} 1 & \text{for } t \geq 0, \\ 0 & \text{otherwise.} \end{cases}$$

Show all your calculations.

Solution for Problem 1(c)

Using (4), we have

$$Y_u(s) = \frac{1}{m\ell^2} \frac{1}{s(s^2 + \frac{g}{\ell})}.$$

By partial fraction expansion, $\frac{1}{s(s^2 + \frac{g}{\ell})} = \frac{\alpha}{s} + \frac{\beta s + \gamma}{s^2 + \frac{g}{\ell}} \Rightarrow 1 = \alpha(s^2 + \frac{g}{\ell}) + (\beta s + \gamma)s$, wherein $s = 0$ gives $\alpha = \ell/g$, $s = 1$ gives $\beta + \gamma = -\ell/g$, and $s = -1$ gives $\beta - \gamma = -\ell/g$. Therefore, $\beta = -\ell/g$, $\gamma = 0$, and we obtain

$$Y_u(s) = \frac{1}{m\ell^2} \left(\frac{\ell/g}{s} - \frac{(\ell/g)s}{s^2 + g/\ell} \right) = \frac{1}{mg\ell} \cdot \frac{1}{s} - \frac{1}{mg\ell} \cdot \frac{s}{s^2 + g/\ell}.$$

Taking inverse Laplace transform to both sides of the above, we conclude

$$y_u(t) = \frac{1}{mg\ell} \left(1 - \cos \left(\sqrt{\frac{g}{\ell}} t \right) \right).$$

(d) The DC gain of a control system is defined as the steady state value of the unit step response, i.e., $\lim_{t \rightarrow \infty} y_u(t)$ for unit step input $u(t)$. Use part (b) to argue in 1 sentence why we can/cannot use the Final Value Theorem (FVT) to determine the DC gain for our pendulum control system.

If FVT is applicable, then compute the DC gain using FVT. If FVT is not applicable, then compute the DC gain, if it exists, by taking $t \rightarrow \infty$ limit of your answer to part (c).

Solution for Problem 1(d)

FVT cannot be applied to $Y_u(s)$ because from part (b), all poles are on the imaginary axis, i.e., on the closed right-half-plane (RHP).

Using our answer in part (c), we find that the DC gain $\lim_{t \rightarrow \infty} y_u(t)$ does NOT exist because the limit does not exist for the $\cos(\cdot)$ term.

(e) In MATLAB documentation, look up the inbuilt commands `ss` and `stepplot`. The command `ss` creates a state space LTI model using the user specified matrices $\mathbf{A}, \mathbf{B}, \mathbf{C}, \mathbf{D}$. The command `stepplot` accepts state space model as input and plots its unit step response within a desired time interval for zero initial conditions.

Using these two commands, write a MATLAB code to simulate the unit step response of the linearized pendulum model over time $t \in [0, 50]$ for $m = 1$, $\ell = 2$, $g = 9.81$. This code will also help you verify your answers to parts (c)-(d).

Solution for Problem 1(e)

See the posted MATLAB file `SolutionPS03p2e.m` in CANVAS. The MATLAB plot indeed confirms our answers to parts (c)-(d).